AttackGraph AI

See what a cloud permission change unlocks, before you deploy it.

AWS AI Fest 2026 | 15 October | Defensive review prototype

One permission change completes a route to admin

CI deploy user
Build pipeline principal

→

Post-build hook
Lambda workload

→

Deployment admin
Protected IAM role

Changed: PassRole false to true. A route counts only when all seven declared conditions hold.

PassRole, create, invoke, code control, Lambda trust, same account and resolved policy controls.

How the prototype works

1 Baseline + proposed snapshots
Hand-authored synthetic JSON. Validate schema, references and declared coverage.

2 Deterministic analysis
Evaluate direct S3 read and role use via Lambda. Build witness routes with NetworkX.

3 Compare + simulate the fix
Find new access. Revoke the new grant on a copy, then recompute routes and expected access.

4 Report + pull-request check
Streamlit, static demo and Markdown output. The CI check fails on new risk or incomplete analysis.

Amazon Bedrock
Amazon Nova Pro via Converse.

Receives aliased evidence after analysis.

Returns explanation text. IDs are validated. It cannot add findings, decide severity or choose a fix.

0 to 1	1 to 0	2 of 2
New high-risk path	After revocation	Expected access kept

Verified in this model. Synthetic data only. Nothing is deployed or executed.

No Terraform/IAM policy parser or live discovery. Broader AWS coverage is future work.

attackgraph-ai.vercel.app/demo